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Contenido

Revisión Modelado de gemelos digitales de incendios forestales: avances de investigación en técnicas de detección, simulación y predicción Yuting Huang 1, Jianwei Li 1,* y Huiru Zheng 2
1 College of Physics and Information Engineering, Fuzhou University, Fuzhou 350116, China; 221120024@fzu.edu.cn 2 School of Computing, Ulster University, Belfast BT15 1ED, Reino Unido * Correspondencia: lwticq@163.com Resumen: Los incendios forestales ocurren con frecuencia en diversas regiones del mundo, causando graves daños a los recursos naturales y humanos. Los métodos tradicionales de prevención y gestión de incendios forestales a menudo se ven obstaculizados por dificultades en la monitorización y una baja eficiencia. La tecnología de gemelos digitales, como modelo de simulación virtual altamente integrado, muestra un gran potencial en la gestión y prevención de incendios forestales. Al mismo tiempo, la combinación virtual-real de la tecnología de gemelos digitales puede proporcionar nuevas soluciones para la gestión de incendios forestales. Este artículo resume las tecnologías clave necesarias para establecer un sistema de gemelo digital de incendios forestales, centrándose en los requisitos técnicos y los avances de investigación en detección, simulación y predicción de incendios. Asimismo, este trabajo propone el modelo de gemelo digital de incendios forestales (WFDT, por sus siglas en inglés), el cual integra datos en tiempo real y simulaciones computacionales para replicar y predecir el comportamiento del fuego. La síntesis de estas técnicas en el marco de un gemelo digital ofrece un enfoque integral para la gestión de incendios forestales, proporcionando información crítica para que los responsables de la toma de decisiones mitiguen los riesgos y mejoren las estrategias de respuesta ante emergencias. Palabras clave: gemelo digital; incendios forestales; modelo de propagación del fuego; detección de incendios; visualización

Introduction

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With global warming and the expansion of human activities, adverse conditions have led to the frequent occurrence of wildfires around the world [1]. Wildfires, as a major global natural disasters, have become one of the serious challenges facing forest ecosystems and human societies globally. Modeling of Wildfire Digital Twin: Research Progress in Detection, Simulation, and Prediction Wildfires not only caused devastating damage to human beings, biodiversity, and ecosystems but also often consume a large amount of human, material and financial resources [2,3]. Techniques. Fire 2024, 7, 412. The "Black Summer" wildfire that lasted for several months in Australia in 2019 [4], the California wildfires in the United States in 2020 [5], and the wildfire in Liangshan, Sichuan Province, China [6], have had a serious impact on the resources of various countries and even the world. Academic Editor: Grant Williamson Received: 13 September 2024 The prevention and control of wildfires has become a global problem. Traditional wildfire management and response methods Revised: 9 November 2024 often face limitations. Therefore, there is a need to introduce new technological tools to improve the efficiency of fire response and suppression and to reduce the damage caused by fires. Digital twin (DT) is a virtual digital copy with a physical entity. Accepted: 11 November 2024 Published: 12 November 2024

It refers to the technology behind the modeling, simulation, and analysis of physical entities, processes, or systems through digital technology to achieve the monitoring, simulation, prediction, and optimization of its real-time operating state [7]. Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. With the development of computer technology, fire spread simulation has become an important tool for wildfire prevention and control [8]. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/). Recent studies have highlighted the inherent complexity and unpredictability of wildfire spread dynamics [10]. Traditional methods of wildfire prevention and control often

Fire 2024, 7, 412 2 of 25

struggle with limited data, slow response times, and difficulty accurately predicting fire behavior [11]. To address these challenges, we proposed developing a wildfire digital twin model using digital twin technology. A digital twin offers a dynamic, real-time virtual representation of the physical environment [12], enabling more precise modeling and simulation of wildfire scenarios. This technology has been shown to enhance real-time situational awareness [13], improve the accuracy of wildfire predictions, and optimize resource allocation during firefighting efforts. The wildfire digital twin model can provide a basis for monitoring, early warning, prediction, approximate simulation, and decision-making in wildfire management [14,15]. It can also support virtual training for fire emergency responders, helping them improve their response speed and capabilities [16]. Consequently, the application of digital twin technology to wildfire management holds considerable theoretical and practical significance. This paper first introduces the definition and fundamental principles of digital twin technology, followed by a systematic review of the relevant technologies required for a digital twin system tailored to the wildfire spread process. Focusing on the entire lifecycle of wildfire spread, we discuss the latest advancements and practical applications in areas such as wildfire monitoring and early warning, fire spread simulation, and digital modeling, providing a detailed categorization of the associated technologies and methods. Through an analysis of various research outcomes, we propose, for the first time, a comprehensive wildfire digital twin model designed to assist related researchers in leveraging digital twin systems for wildfire management. The advantages and challenges of digital twin technology in wildfire management are also analyzed. This review aims to provide valuable insights and guidance for researchers and decision-makers, thereby promoting the broader application and development of digital twins in wildfire prevention and response.

Materials and Methods

2.1.

Data Collection and Analysis

To establish the research background and assess the current status of studies related to the topic of our review, we conducted a comprehensive literature analysis in the Web of Science (https://www.webofscience.com/wos/, accessed on 12 July 2024) database. Common wildfire types typically include forest fires, grass fires, crown fires, and bush fires [17]. Studies were selected based on the inclusion of specific keywords related to wildfire management and digital twin technology. So, we established the following search keywords in this database: “forest fire”, “wildfire”, “wildland fire”, “grass fire”, “crown fire”, “bush fire”, and “fire”. Then, we combined these keywords with “digital twin” for an advanced search. Since digital twin is a relatively new technological concept, the data collection for this study primarily focused on the past five years. Some early, sporadic foundational

research may have been excluded, but after reevaluation, this was found not to affect the comprehensiveness of our trend analysis. Only peer-reviewed journal articles, conference papers, and reviews were included to ensure the quality and reliability of the sources. Thus, the data in Figure 1 were obtained.

The results in Figure 1 show a growing trend in the number of research papers related to the application of digital twin technology in wildfire and fire studies over the past five years. Notably, the number of papers on “fire” combined with “digital twin” has steadily increased from 10 in 2020 to 36 in 2024.

In contrast, studies specifically focused on “wildfire” have also risen, though at a slower rate, starting from 3 papers in 2020 and reaching 12 papers by 2023.

This trend suggests an increasing interest in integrating digital twin technology into fire management research, highlighting its potential value.

However, there are still fewer studies on “wildfire” compared to the broader category of “fire”. This suggests a gap in research focused on applying digital twin technology to wildfires.

The increasing trend emphasizes the importance of developing more refined digital twin models tailored specifically to the unique characteristics and challenges of wildfires, such as rapid spread and unpredictable behavior.

Fire 2024, 7, x FOR PEER REVIEW 3 of 26
increasing trend emphasizes the importance of developing more refined digital twin models tailored specifically to the unique characteristics and challenges of wildfires, such as rapid spread and unpredictable behavior. From this data, we can see that there are still obvious theoretical and technological gaps in the research of digital twin for wildfires. The purpose of writing this review is to fill this gap by gathering current research and providing new directions to relevant researchers.

gathering current research and providing new directions to relevant researchers. FFF iii ggguu urr ree e 11 1.. . TT Thh hii iss si si i s sa a afi figiug g uruer r e eo fo of f h tet h hne eu n nmu ubm merb b eoe r rf po oaf f pp pea arp pse error s sla r rte eel lda a t tte eod dt h t teo o fit t h hele ed fi fioe efl ldd di go oif ft ad dli itg gwi i t tia anl l. t tWw wei i n nu. . sW Wede e a u uds sve ead dn ca aed dd- - vanced search to separately retrieve papers using keywords such as “wildfire” and “fire” with the search engine. Data include studies published until August 2024. “todpigit aolf t”wdiing”it.aDI tawtainin”c. IDuadtea sitnucdluiedsep sutubdliisehse pduubnlitsilhAedu guunstilt2 A0u24g.ust 2024. 22..22.. DDaattaa OOVveerrvviieeww 2.2. Data Overview AAss wwee hhaavvee ffoouunndd,, tthheerree iiss lleessss rreesseeaarrcchh oonn tthhee aapppplliiccaattiioonn ooff ddiiggiittaall ttwwiinn iinn wwiillddffirreess.. As we have found, there is less research on the application of digital twin in wildfires. AAnndd tthhee ffeeww eexxiissttiinn aapppplliiccaattiioonn ooff ddiiggiittaall ttwwiinn iinn ccoonnjjuunnccttiioonn wwiitthh fiffirreess iinn tthhee reelleevvaanntt And the few existing applications of digital twin in conjunction with fires in the relevant literature often lack comprehensive integration [18–20]. In fact, through our research on the relevant literature [21], the digital twin has been found to have

great potential and aadndvaandtavgaenst aing efisrien mfiarenamgaenmaegnetm (Feingtu(rFei g2u).r len2 t)h.el nfiethlde ofife eldnvoifroennvmireonntaml emnotanlitmoroinnigt,o sruincgh, advantages in fire management (Figure 2). In the field of environmental monitoring, such asus cwhiladsfiwrei lmdafinreagmemaneangt,e dmigenitta,l dtwigiinta tlectwhninolotegcyh nenoalobgleys erenaalb-tliemser emalo-ntiimtoerimngo onfi teonrvinirgono-f as wildfire management, digital twin technology enables real-time monitoring of environ- menevnirtoaln cmoenndtiatllocnosn d[2it2i]o. nHso[2w2e].vHero, wduueev etro, tdhuee ctoomthpelecxoimtyp alenxdi tyaaarniadbvilaitryia obfi ltihtye ofofrtehset feonrvesi-t mental conditions [22]. However, due to the complexity and variability of the forest envi- reonnnvmiroenntm aenndt tahned htihghe hsuigdhdesundnedsesn onfe fissreosf, tfihreerse, itsh aer reeliastaivree llaactikv eofl arecskeoarfcriche sreeasruchts roensu thltes ronment and the high suddenness of fires, there is a relative lack of research results on the aopnptlhiceaatpiopnl iocfa tdioignitoafl dtwigiint atlot twhein fiteoldth oef fiweillddfiorefews.i IPdrfierveiso.uPsrlyev, tihoiuss tleyc,htniosisltoegcyh nwoalso gmyowstalys application of digital twin to the field of wildfires. Previously, this technology was mostly umsoesdt liyn uthseed fiienldth oef fifierled foorf ifinrdeoforr ibnudilodoirnbgu fiirldesin. gfires. used in the field of fire for indoor building fires. FFiigguurree 22.. TThhee ppootteennttiiaall ooff ddiiggiittaall ttwwiinn iinn fifirree mmaannaaggeemmeenntt.. Figure 2. The potential of digital twin in fire management. Intermsoffireprevention,artificialintelligenceiswidelyappliedinfiresafetyand hasbecomeakeyenginedrivingthedevelopmentofdigitaltwinintelligentfireprotection. Ding et al. proposed an intelligent emergency digital twin system based on computer visionanddeeplearning. Theevacuationdatageneratedbythesystemhelpsevacuatefire scenes,achievingdetection,tracking,andprivacyprotectionofevacuees[23]. Zhangetal.

Fire 2024, 7, x FOR PEER REVIEW 4 of 26 In terms of fire prevention, artificial intelligence is widely applied in fire safety and has become a key engine driving the development of digital twin intelligent fire protec- Fire2024,7,412 tion. Ding et al. proposed an intelligent emergency digital twin system based on comp4uoft2e5r vision and deep learning. The evacuation data generated by the system helps evacuate fire scenes, achieving detection, tracking, and privacy protection of evacuees [23]. Zhang et al. pprooppoosseedd aa nneeww AArrttiificciiiaall llnntteellliiggeennccce DDiiggiittaall FFiiirreefifigghhttiinngg ((AAIIDD--FFiiirree)) ffrraammeewwoorrkk uussiinnngg AAl eennggiinne--ddrriivveenn ffoorr rreeaall--ttiimnee rreeccooggnniittiioonn ooff ccoommpplleexx bbuuillddiinngg fifirree iinnffoormmaattiioonn,, iinncclluuddiinngg fifirree ddeevveellooppmmeenntt,, fifirere spspreraeda,d a,nadn fidrefi rmeomveomveemnte [n2t4][.2 l4n] .terlnmste orfm fisreo rfefispreonrsees,p tohne sdei,gtihta el dtwigiint aclatnw siinmcualnatsei mfiuel astceenfiarreiiossc einn arreiaols tiinmree, ainlctilmudei,nign cthlued sipnrgetahde ospf trheea dfiroef, tthhee tfiyrpee,sth oef tcyopmebsuosfticbolmesb, uanstidib tlhees ,dainstdribthuetiodnis torfi bsumtiookne,o tfos pmroovkied,et odepcriosvioidne sudpecpiosriot nansdu pcpomormt aanndd cpolamnms afonrd fiprlaenfisghfoterrfiser e[2fi5g]h. tlenr sin[2t5e]r.alcntivinet efirraec tisvuepfipreesssuiopnp rseismsiuonlatsiiomnu, lMatieonng, Mete anlg. ectoanl-. csotrnuusstreudc tdeidffderifefnetr ecnotmcboumsbtiubsleti bmleodmeolsd aenlsda snudccsuescscfeuslsfyu vlliysuvaisliuzaeldiz tehde tohcecuorcrceunrcreen acneda enxd- etixntginugisuhisinhgin bgebheahvaiovri oorf ofofrfeosrte fistrefisr eins i3nD3 sDcesnceesn [e2s6][2. 6P]r.oPvridovinidg ian gfoaunfoduantidoant ifoonr tfhoer tahpe- applipclaitciaotnio onf owfwildilfidfier edidgiigtaila tlwtwini.n D. Douourvravsa sete taal.l .ccoommbbinineedd cceelllluullaarr auuttoommaattaa aanndd ddiiggiittaall ttwwiinn ffoorr tthhee fifirrsstt tiimnee aanndd pprooppoosseedd aa ddiiggiittaall ttwwiinn pplaattfoormm too mmoonniittoorr tthhee hhuummiiddiitty aanndd tteemmppeerraattuuree ccoonnddiittiioonnss iinnssiiddee aa bbuuillddiinngg tthaatt ppreeddiicctss aanndd ssiimuullaatteess tthhee spprreeaadd ooff fifirree wwiitthhiinn aa bbuuillddiinngg [[2277]]..

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and full-size tunnel fire tests, demonstrating the feasibility of using 3D environments and full-size tunnel fire tests, demonstrating the feasibility of using 3D environments and digital twin in real-time fire safety management [28]. The “Wildfire Digital Twin” project, digital twin in real-time fire safety management [28]. The “Wildfire Digital Twin” project, planned by NASA for 2024, is currently the most comprehensive application combining digital twin technology with wildfire management [29]. It represents a major step forward in this field. The digital twin system built in this project will use artificial intelligence and machine learning to predict potential burn spread paths in real time, merging data from multiple sensors to produce global models with high precision. It will provide a valuable advanced tool for firefighters and fire managers. Figure 3. The overall digital twin framework for tunnel fire safety management [28]. Figure 3. The overall digital twin framework for tunnel fire safety management [28]. 33.. DDiiggiittaall TTwwiinn TTeechhnnoollooggy 33..11.. DDeeffinniittioonn ooff DDiiggiittaall TTwwiinn TTthee tteerrmm ““DDiiggiittaall TTwwiinn”” wwaass nnoott ffoormmaalllyy iinnttroodduceedd uunnttiill 22001100 iinn aa NNaaattiioonnaall AAeerroo-- nnaauuttiicss aanndd SSppaacee AAddmminisitraratioonn (N(NAASASA) t)ectehcnhicaicl arelpreoprt [r3t0[3. 0N].ANSAA dSAefidneefidn ite ads iat saisma- suilmatuolant iponropcreoscse tshsath fautfluy luytiulitileizs epshpyhsyicsaicl amlmodoesl, ss,esnesnosrosr,s o,oppeeraratioonnaal lhhisistoorry,, aanndd ootthheerr ddaattaa too iinntteeggraatee mmuulltii--ddiisscciipllinaarryy,, mmuulltii--pphhyssiicss,, mmuulltii--ssccaallee,, aanndd mmuulltii--pprroobbaabbiillitty [[3311]].. SSiinncee tthheenn,, ddiiggiittaall ttwwiinnss hhaavvee bbeegguunn too hhaavvee aann iimppaacctt iinn bbootthh aaccaaddeemmiiaa aanndd iinndduussttryy.. Related concepts such as cyber–physical worlds and the metaverse are also beginning to be widely disseminated and applied by scholars in a variety of research fields and industrial applications [32], such as industrial manufacturing, architecture and planning, and aerospace [33].

3.2.

Technical Framework of Digital Twin

The core of digital twin lies in creating one or more highly detailed virtual models that reflect the state of physical entities, which are fully synchronized with their physical ver-

Fire 2024, 7, x FOR PEER REVIEW 5 of 26 industrial applications [32], such as industrial manufacturing, architecture and planning, and aerospace [33].

3.2. Technical Framework of Digital Twin

The core of digital twins lies in creating one or more highly detailed virtual models that reflect the state of physical entities, which are fully synchronized with their physical svioenrssiio[3n4s] .[3T4h]i. sTshyins cshyrcnohnrzoantiioznatiisobna isse bdaosneds eonnns osersnsaonrds oatnhde rodthaetar dacaqtau iascitqiuonisitteicohnn toelcohgnieosl,- woghiiecs, twwhheoicrhe tihcaeollryectiocnaltliyn ucoonutsilnyumouosnliyt omrorenaitlo-tri mreeald-taitmaea bdoautat apbhoyusti cpahlyosbijceactl osb(pjeocstsit (iopno,- tseimtiopne,r atetumrep,ehruatmuride,i thy,upmreidssiutyr,e ,pertecs.s)uarned, etrtca.n) samnidt tthraensesmdaitt athbeascek dtoattah ebadcigki ttaol tmheo ddeilg.iltнал amdodditeiol.n l,nd aigdidtaitlitown,i ndsigailtsaol itnwciInusd ealasod aintaclaundael yas disaftraa maneawlyosrisk ffoarmperwocoersks ifnogr ipnrpouctesdsaintag ainndpusti mduatlaa tainngd psrimeduiclatetidngfu pturerdeiscttaetde sf,uatlulorew sintagtefso,r amllowweinefgf efcotri vmeodreec iesffioenct-imvea kdiencgisaionnd- pmreadkiicntgiv aenmd apinretednicatnivcee [m35a]i.nDterniavneceeb [y35d]a. tDa,rifvraemn ebwy odraktsa., afrnadmmewodoerlkss[, 3a6n]d,d migoitdaelltsw [3in6s], canmonitor,simulate,predict,andoptimizetheirphysicalentities. digital twins can monitor, simulate, predict, and optimize their physical entities. In this paper, the composition of the digital twin framework is divided into the In this paper, the composition of the digital twin framework is divided into the fol- followingthreeparts: physicalentity,digitaltwincoreentity,anduserentity,eachwith lowing three parts: physical entity, digital twin core entity, and user entity, each with dif- differentfunctions(Figure4). Thephysicalentityisthecornerstoneofthedigitaltwin. ferent functions (Figure 4). The physical entity is the cornerstone of the digital twin. The Thedigitaltwincoreentityisthekeypartofrealizingthevirtualizationandsimulationof digital twin core entity is the key part of realizing the virtualization and simulation of physicalentities. Additionally,theuserentityisthefinalapplicationlinkofthedigitaltwin. physical entities. Additionally, the user entity is the final application link of the digital Digitaltwintechnologyorganicallycombinestheabovethreeparts,coveringthecomplete twin. Digital twin technology organically combines the above three parts, covering the processfromdataacquisition,processing,andanalysisistouserinteraction,andrealizing complete process from data acquisition, processing, and analysis to user interaction, and thedynamicmappingofphysicalentitiesanduserentities. realizing the dynamic mapping of physical entities and user entities. Figure 4. Technical framework of digital twin. Figure4.Technicalframeworkofdigitaltwin. 3.3. TheApplicationofDigitalTwin In the process of continuous development and maturity of the technology, digital twinhasbecomeanadvancedvirtualsimulationandpredictiontool. Itisnowgradually expandingtoamixtureoffields,includingmanufacturing,energy,healthcare,andsoon (Figure5).

Fire 2024, 7, x FOR PEER REVIEW 6 of 26

3.3. The Application of Digital Twin

In the process of continuous development and maturity of the technology, digital twin has become an advanced virtual simulation and prediction tool. It is now gradually expanding to a mixture of fields, including manufacturing, energy, healthcare, and so on (Figure 5). In Industry 4.0, digital twin is used to improve productivity and reduce costs through the real-time monitoring of production lines, predictive maintenance of equipment, and simulation of manufacturing processes [37]. In addition, the technology is widely used in product design, production, forecasting, and cycle management [38,39]. Digital twin is widely used to enhance urban infrastructure management [40], opti- mize traffic flow [41] and emergency response and management [42], and assist in smart city management and improve the sustainability of cities [43–46]. In medical care, digital twin combined with related technologies provide new solu- tions for patient care, predictive analysis, and clinical operation training [47–49]. In aerospace, digital twin helps predict and solve potential problems by simulating the life cycle of aircraft, from design to manufacturing, use to maintenance [50,51]. In ad- dition, digital twin

technology is widely used for satellite systems, spacecraft, and lunar exploration [52]. Digital twin enables the monitoring, prediction, and optimization of energy and environmental systems by creating digital copies of physical entities [53]. Digital twin effectively monitors air quality [54] and water resources [55], predicting environmental change trends through real-time data collection and virtual model simulation. In ecosystem management, digital twin helps to understand the dynamic changes of ecosystems and develop more effective conservation and restoration strategies [56,57].

4. Wildfire Detection and Real-Time Data Acquisition

The development of a wildfire digital twin relies heavily on advanced methods and technologies for real-time monitoring and data collection. This section discusses key advances in wildfire detection and real-time data collection methods to highlight their role in the wildfire digital twin model.

4.1. Wildfire Detection

To reduce losses from wildfires, countries around the world attach great importance to wildfire detection, which is an essential part of wildfire management. The initial phase of fire management involves the timely and accurate detection of fires, and once detected, the life cycle of aircraft, from design to manufacturing, use to maintenance [50,51]. In continuous monitoring is essential to assess their development and potential threat. The addition, digital twin technology is widely used for satellite systems, spacecraft, and lunar exploration [52]. Digital twin enables the monitoring, prediction, and optimization of energy and environmental systems by creating digital copies of physical entities [53]. Digital twin effectively monitors air quality [54] and water resources [55], predicting environmental change trends through real-time data collection and virtual model simulation.

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4.1. Wildfire Detection

To reduce losses from wildfires, countries around the world attach great importance to wildfire detection, which is an essential part of wildfire management. The initial phase of fire management involves the timely and accurate detection of fires, and once detected, continuous monitoring is essential to assess their development and potential threat. The real-time data provided by wildfire detection is an important input source for the wildfire digital twin model.

Accurate and effective detection data can ensure that the digital twin model truly reflects the real-time status of the fire scene. The measures for wildfire detection can usually be categorized into four spatial levels, namely, ground patrol, near-ground observation, aerial patrol, and space satellite monitoring [58]. Various technologies, such as IoT sensors, remote sensing, and drones, are

iag all i-- ttt aii lmm twee dd inaa tt maa opp drroo evv l.i i Addee cdd cu bb ryy a tww e ii alldn d dfifir r eee ff dd eee ctt tee ivcctt eii oo dnn e tii ess c aa tnn io ii nmm dpp aoo trr att aa cnn antt i i enn npp suu utt r ss eoo tuu hrr acc tee t ff hoo err dtthh igee i ww taii lll dd twfifi irr nee real-time data provided by wildfire detection is an important input source for the wildfire mddii ogg diitt eaa lll ttt rww uii lnn y mm reofl odd ecee tll s.. AA thcc ecc uu rerr aaa ltt -ee t iaa mnn edd see tfff atee ucc sttii ovv fee tdd hee ett ee ficc rtt eii oo snn ce dd naa ett .aa ccaann eennssuurree tthhaatt tthhee ddiiggiittaall ttwwiinn digital twin model. Accurate and effective detection data can ensure that the digital twin mmooddee Tll h tt erru u mllyy e arree sflu fl ree ecc stss f otthh re e w rr iee laa dll fi--tt rii emm dee e ss ttt eaa ctt tuu ioss n oo ff c atthh nee u fifi srr uee a ss llcc yee nn bee e..

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nn e tii ess c aa tnn io ii nmm dpp aoo trr att aa cnn antt i i enn npp suu utt r ss eoo tuu hrr acc tee t ff hoo err dtthh igee i ww taii lll dd twfifi irr nee real-time data provided by wildfire detection is an important input source for the wildfire mddii ogg diitt eaa lll ttt rww uii lnn y mm reofl odd ecee tll s.. AA thcc ecc uu rerr aaa ltt -ee t iaa mnn edd see tfff atee ucc sttii ovv fee tdd hee ett ee ficc rtt eii oo snn ce dd naa ett .aa ccaann eennssuurree tthhaatt tthhee ddiiggiittaall ttwwiinn digital twin model. Accurate and effective detection data can ensure that the digital twin mmooddee Tll h tt erru u mllyy e arree sflu fl ree ecc stss f otthh re e w rr iee laa dll fi--tt rii emm dee e ss ttt eaa ctt tuu ioss n oo ff c atthh nee u fifi srr uee a ss llcc yee nn bee e.. categorized into four spatial lev- model truly reflects the real-time status of the fire scene. els, nTT ahh mee e mm lyee , aa gss ruu orr ueess n dffoo rr p ww atii rll odd lfiifi , rr nee e dd aree -tt gee rcc ottii uoo nnn d cc aa onn b suu ess ruu vaa all tll iyy o nbb ,ee acc eaa rtt iee agg l oo prrii azz tee rdo d l ,ii nn att noo d ff oo suu prr a ss cpp ea a sttii aaa tll e lll lee ivv te-- The measures for wildfire detection can usually be categorized into four spatial lev-meellss o,, n nn itaa omm rien ell gyy ,, [5gg 8rr]oo . uu Vnn add ri opp uaa stt rr too ell c,, hnn nee oaa lrr o-- ggg irr eoo suu , nn sudd c hoob b ass see rr lvv oaa Ttt ii soo enn n,, s aa aee rsrr ,ii aa rll e mppaa ott trr eoo ll s,, e aa nnn sidd n gss ,pp aaa nccee d ss daa rtt oee nllll eiitt see , Fire2024,7,412 els, namely, ground patrol, near-ground observation, aerial patrol, and space sat7elolfit2e5 amm reoo nn eii mttoo prrii lnn ogy g e [[d55 88 t]] o.. gVV aaa trr hiioo euu r ss h ti t geecc hhh -qnn uoo alloo ligg tyiiee ss d,, a ss tuu a.cc hh T aaa bss l ell oo 1TT s sh see onn wssoo s rr tss h,, err ee dmm etoo ett cee t iss oee nnn sm siinn egg a,s , uaa rnn edd s dd anroo dnn ree ess -, monitoring [58]. Various technologies, such as IoT sensors, remote sensing, and drones, laa arr tee e deemm tepp clh loo nyy oee ldd o gttoo ie sgg aa fotthh r ee thrr ehk fiigo ghu h-- rqq suu paa all tii itt ayy l dd leaa vtt eaa l.. s TT : gaabb rollee u n11 dss hh poo aww trso s ltt ,hh nee e add ree -tt gee rcc ottiu ioo nnn d mm obeeaa sessur u vrr aee tss i oaa nnn ,dd a erree r--- are employed to gather high-quality data. Table 1 shows the detection measures and re- ill aaa ltt ee pdd a tt ter e occhl h , nn aoo nll doo gg sii pee ass c ff eoo rr s attht h eee l fff ioo teuu rm r ss opp naa itt tii oaa rll i ll nee gvv .ee llss:: ggrroouunndd ppaattrrooll,, nneeaarr--ggrroouunndd oobbsseerrvvaattiioonn,, aaeerr-- leamtepdI oteyehdntoologgaitehse frohr itghhe -fqouuarl sitpyadtiaatla l.evTealbsl:e g1rosuhnodw psatthreold, enteeacrt-igornomuneda soubseesravnatdiorne,l aateerd- iiaall ppaattrrooll,, aanndd ssppaaccee ssaatteellllittee mmoonniittoorriinnngg.. itaelc phnatorloolg, aiensdf ospratchee sfaotuelrlstpe amtioanlilteovreinlsq:. ground patrol, near-ground observation, aerial Table 1. Summary of wildfire detection measures and technologies. patrol,andspace satellitemonitoring. TTAabblllee 11.. SSuummmaarryy ooff wwiiillddffirree ddeetteeccttiioonn mmeeaasssuurreess aanndd tteecchhnnoollooggiieess.. Table 1. Summary of wildfire detection measures and technoloSgcieosp. e of S SS p pp a aa t tt i i aa l ll L LL e ee v vv e ee l ll s ss Table M MM 1 e e . e a a S a s s u s u u m r r r m e ee s ss a ryofwildfi R R r R e e ee d l ll a e aa t t t e e ee c d dd ti o T TT n e ee c cc m h hh e n nn a o oo su l ll o oo re g gg s y yy a ndtechnoAlo S Sp Sgi c cp ce o oo lsi p p.p c e ee a t o oo io f ff n S SS k kk e

ee t t t c c c h h h M M M a a a p p p Spatial Levels Measures Int R e r e n l e a t t e o d f T T h e c
in h g n s o l o g y A A p p p p l i c c a a t t i o o n n Sketch Map Spa G t i r a o l G G G G u L n r r r e o o o o
d v u u u u e n n n n l s d d d d 1 2 . . M G 1 2 1 2 1 2 1 2 r a . . o M n u u n e a M G M G s
M G s M G s a d l t t t s r a r a r a r p m u a a a a o o o t t t a n n r n n i i u o u u u e t o o o u u u r n
s n n n n n n o a a a a i d d d d l t l l l o p p p p r i m m m m a a a a n t t t t g o o o r o r r r o o o n n n n
l l l l i i i t t t t o o o o (l n r r r r l n i i i e i o n n n t n t T e w g g g r) n l (w / (w n l (w l (w o r l
n l l n l n e W r i o o o o e o o o c o t k t t t l T T e T e T e a r i r r s r o a r r r k r k k / t) k)) f t e n n
i / n / / e s s W s s l o T W W W W e e d e / e / / n h W W W s W t i t t r i i T s i i i r o r r o e n r o e i
i i s i e e e e l f g f r r f r r c e e l l l l e e e e s h T n s T e e e T e l l l s n s s s s s s h e e h e h e o s s
s s o s s s s i i r n n s l s n s s s s s s o e e e g e g g c c c c g n n n n s o s o o s o y s s s s m m m m
o o o o r r m m m r m n n n n u u u u e e e e - - - t t t t - - - S E S E S E F S E F S E F m m m a m
i i m a i a a a x x x s S s s s s a i e e a a e a a i i i c l l l l d d l y d l l l l o y y y l l l f p a l l f f o l f f a
A a a a o o e o c o o o o r c c c c c c c e c r r r p o e c c c c a a s e e a e e f s e e p e e t s t s t s s
A s s s i s i a t t l t t i o o o s s s s i s b r p a a a a c i i n i n i e n l b b b b r r r p a r e a s s s e e e l l l t
s l r e e e a i i a a a e c o s s s s g r r r a n e e e e i t o i g g g g o n i i i n o o o s o n n n n s s s s
SketchMap s 1 t a . t i o n s W t a t t i c o h n t o w e r s c o m m n A n n u i i u c c n a t a a i o t t c i m i
a o o o t n n i n a o t n e d S y s t e m s / l n - F F i i x x e e d d l o l o c c a t i t o i n o s n s N e a N N N N r e e e e G a a
a a r r r r r o G G G u G n r r r r d o o o o u u u u n n n n d d d d 1 2 3 . . . W S U 1 2 2 3 1 2 3 1 2 3 u
n a r t m v c e h a i W S S U v W S U n W S U l t l o u u u n e u a n n a a a w h e n r r
r r m m m t t t d v v v i c e c c c e e e e e r h h a h a a a l s i i i e c e n n n l l l t t a t l s l l r l n n o n
o o a a a a i m a (w e n w e n n w e n U l e d d d c c c c e e e A e e e e a r r V s s s c c c c a a a a)
m m m m a a a e e A a e e e e e n r r u r r r r i i d a a a a a a t o l l l v A f f i A f A f m m r i r r s
u a u a u a a i a r b r r t t t t e e o e e o e g o l e d d d d d m m i m n l S i a a a a a g a g y n n n
t t t h / s e e e d d d d t t h d d d e i v m v v m i S S g i i i s y s a s y s y s h / i i g i s s s b b b b l i t r
t t n n l l l e l e e e e e e f g m m m s r / l l o l l a i i i s s s r l g g g g / / u e / l h l h h l h d t n n n t t t i t
- - o - n L L C L C L C L C a a a o a a o o o r r r m r r m m m g g g g g e p e e e e p p p l f e l l o f l f f f
e e e o o x o o r x x e r r r t s e e e e t t t t s s r s s e e e r a t t t t r r a r r r a a a a i e a a n r r r a i i i
e e e e s s n n n a a a a s s s s s s 3. U n m a n n e d a e H r i i g a h l i m r m e s a a o g g l u i n t t i g g o / n H H i i g g h h
r r e e s s o o l l u u t t i o o n n C o m p l e x t e r r a i n s v 1 e . h i c l e H v v s e e e (h h U l i i c c A c o l l V e e p s s) t (e (U U
r s A A V V)) i R m e a a l g - i T n i g m / e H i i m g h a g r e e s t o r a l u n t s i - o n L a r g e f o r e s t a r e a s v e h i c l e s (U A V)
A e r i a l 1. H 2 1 1 e . . . l i c o A H H p t i e e e r l l r c i s c r c a o o f p p t t p e e r a r s s t r o l R e a l - m R R T i e e m i a s a
s l e l - i T o T i m n i i m m a / l g e n e e f i i m m a r a a e g g d e e a t t r r n a a d n n s v s - i s - L H L L a a a a r r r g r g g
d e e e - f t o f f o o r - e r r r s e e e t s s a a t t c r a a h e r r e e r s e a a g s s i o n s 1. H e l i c o p t e r s R e a l - T i m e
i m a g e t r a n s - L a r g e f o r e s t a r e a s A e r A A i a l e e r r i a a l l 2. A 3 2 2 i . . . r c r a H A f a t i i i p r r g a c c
h t r r r - a a o a f f l t t t i p p t a u a t t d r r e o o l l U A t r V a n s s i m m m b l i i s e s s s s l i i i o o g n n n h / / t l l l
n n n i m f f f r r a a a r g r e e e i d n d d a g n a a d n n d d v v i i s s - H C H H o a a a r m r r d d d - p - t - o t t l o e o - r
x - e r r a t e e e c a a h r c c r h h r a e i r r g n e e i s o g g n i i o o s n n s s A e r i a l 3. H 2 3 3 i . . . g h - a A H H H l t
i i i i r g g t g u c h h h r d - - a - e a a a f l l t l U t t t i p i i A t t a u u u V t d d d s r o e e e l U U U A A A v i
V V V s i s s s b l m i R i i e b b b e l l i l l e i m e s e g s l l h l i o i i i o t g g g t n e i h h h m / t t s t l a e n i i i g
m m m n f i r s n a a a a i g g n g g r e i g i n n n d / g g G g a e n o d - v i s - H C C C C o o o o a m m
m m r d p p p p - l t e l l o e e e x x x - x t r e t e t t r e e e a r r r r c a r r r h i a a n i i s r n n n e s s s g i o n s
S p a S S S c S e p p p p a a a a c c c c e e e e 1. S 1 1 1 1 a . . . t e l l i t S S S S e a a a a m t t t t e e
e e o l l l l l l l n i i i t t t t t t e e e e o m m m m r m i n o o o o g n n n n i i i t t t t o o o o r r r r p R i i i i n
n n n n o e f g g g s g m o i r t o R g t g t s m R g t i R g t o e e e e t y r r r r e e e e n m a m m m a a
a s a m m m t i s t p p p p i n / / / e / e o G G G o o h G h g h o h n m n t t i i i s i l l l s l e e c c e
c c s i o o o / o y n y r b b i b s i s b s i s a g n s n n n e t e e a a a a t d / e f f f n f n e n l l l l m o o o
G o a m s s s p p p p r r r r r i e i / i m m m o o / o m n n o n o r s s G s s g g g a g a a a i i i d / r / l t
t t t t t t t t G o G a G i i i a i i i o o o o o p o b o r e e e n n n n a n h n n n o o o l i i i i - n - c n n - s n s
s s y y y y g g g g s s s s - - - G L G L G L G L G L a a a a l l l l o r o o o o r o r r r g b g g g g b b b b
e a e e e e a a a a l f l l l l o f f f f c o o o o c c c c o r e o o r o r o r v s e e e e v v v v e t s s s s r e e
e e a t t t t a r r r r r g a a a a a a a e e r r r r g g g a g e e e e s e e e e a a a a s s s s
s s y s s t t e e m m / r r a a d d a a r r s y s t e m / r a d a r 4.1.1. U n m a n n e d A e r i a l V e h i c l e D e t e c t i o n 4.1.1.
U n m a n n e d A e r i a l V e h i c l e D e t e c t i o n 44..11..11..U U A n n V m m d a a n n e t n n e e c d d t i

oAAneerriisaatl hVVeעהhdiieccplleelo DDyeemtteeeccnttiitoonno f UAVs to forest sites, using equipped infrared 4.1.1.U UAnVm danetneecdti oAne riisa lt hVee hdiecplelo Dyemteecntito no f UAVs to forest sites, using equipped infrared s s e e n n s s o o UU r r s sAA, , VVc c a a m dd meee ettr ee r a ca cs tt s , ii ,ooa an nn n d ii dsso t otth hh t e heer eddd r eee dppv elli ov oce yy ic s mm e t s o ee nn t c ot at p oo c t affu pUUr t e uAAr rVVe e a ss l r - te tt ooi a m lff -oo t e irr m d eee ss e t tt a dssil ii e s tt tee a o ss i f , , l t suuh so se ii fnnfi tggr h e eees qqc fiu eu r n ii eppe p .p sc U ee eddA n V eiinn . s ff Urra aal A l rro Veew dd s assfo ee lln rn os hs wooiUg rr fs A sh o,- V r a cc h l aatd mi mi g teu he t e ed rr - c aa aet sl s iv t, , o i i ta n ae unnw d i dd ss e ot oo v hr tt ih e h e c ew eldo rr s e s dd ep oee lv r o vv i c ye ii lccwm oee sssse e no tt voo tf i to ecch wa f ae pp U s fi tt uu Ar o e rf r V ees t s h c rre e t eeno aa fi e ll f r . -- o tt e T ii r mm s he c se ee e t a ndd sd e i ee tv .t e ta Taa sn ii h , ll t ss e ua sg oo a ie dff n v i t g ts hh a t e e e n h q tfifie a u rr g ai ee p e b pi ss i l scviet ee d t y nn h t e i e e on .. a fa UU r b ca iAAcr l e i e VV t sd ysss sheingsho-raslt, ictuamdee,rraesm, aontde, oatnhderh daezvaricdeosu tso acraepatsu,rper orevaidl-itnimgeh digehta-qilus aolfit ythae firirael sficreeneim. UagAeVrys taa olll oo aww cc effos o srr hh hii igg gh h-- -aa all ltt tii itt tuu udd dee e ,v v rii eee mww oss too err , cc all noo dssee h vv aii zee aww rdss ooo uff stth h aee r efifi arr see , ss pcc ree onn vee i.. d TT inhh gee aa hdd igvv haan- n qtt uaagg aee li tii yss tt ahh eee r iaa abb l ii fillii rtty e aalnldwh ofotsrp hoigthm-aalptist.uTdhe evcieuwrrse nort icnlonsoev vaiteivwesa opfp throea fichret socwenield. fiTrhee pardevvaennttaigoen iiss tthhee aubsielitoyf itt moo aaa gcccc eee ryssss a hh nii dgg hh h-- oaa tll stt piitt ouu tdd mee,, a rr pee smm . Too httee e,, caa unn rdd re hh naa t zz inaarn rdd ooo vuu ast s i vaa err ee aaa pss p,, rpp orr aoo cvv hii dd toiinn wgg ihh ldiigg fihh re--qq puu raa ell viitt eyy n taa iee orr niia a ill s fifi thrr eee tUoA aVccseesqs uhiipgphe-daltwitiuthdea, drveamnocteed, afinred dheatzeactidoonums oardeealss,, pSaroyvdiidrainsgu lhovigihch-qeutaalilt.yp raoeprioasl efidrea uiimm seaa gg oee f rr Uyy Aaann Vdd s hh eqoo uttss ipp poo ptt e mm d aa wpp iss t.. h TT ahh dee v cc auu nrr crr eee dnn tt fi iinr n enn oo dvv eaa tettii cvv tiee o aa npp mpprr ooo daa ecc lhh s .tt oo S aww yii dlldd irfifi arr see u lpp orr vee ivv cee hnn ettii too ann l .ii ss p tt rhh oee - itmaraggeetrdye atnecdt ihoontsmpootd melacpuss.t Tohmei zceudrrfeonrt tihnenouVvAatViviem aappperroyacehn vtoir wonilmdfienret, pwrehvicehnteionnh aisen tchees puu ossee s eoo dff aUU tAA arVV gss e tee qq duu eit ipp ecpp tee iodd n ww miitt ohh d ae a ldd cvv uaa snn tcc oee mdd i zfifi err dee fdd oee rtt ee thcctt eii oo Unn A mm Vo o idm dee allsg s.. e SS ryaa yy edd nii vrr iaa rss ouu nll moovv eii ncchh t, weett h aa ill c.. h pp err noo --- uthsee eofff eUcAtiVvesn eeqsusiopfpfiedre wdiethte catdiovnan[c5e9d]. fiZrhe adnegtedcteivoenl ompoeddealss. pSeacyidaliirzaesdulUovAicVh seyts atel.m prfoo-r hpp aoo nssee cdd es aa t httaa err egg ffeett e cdd tee ivttee ecc ntt eiioo snn s omm f oo fidd reee ll d cc euu tesstt coo timm oniizz [ee 5dd 9]ff .oo Zrr htthh aen e gUU dAA eVV v eiimm loap agg eee drr ya y see pnn evv cii irr aoo lnn izmm edeenn Utt,, A ww Vhh siicy chh st ee enn m-- pwoisldefidr ae tdaertgeectt idoent,ebctaisoend monodwehl iccuhestZohmaiozedde sfiogrntehde aUsAalVie inmcyaagdeertye cetniovnirmonemtheonndt,t wohenichha nence- fhh oaa rnn wceee iss ld tthfi her e e ee dffff eee tcc ett ciivv tiee onn nee ,ss bss a oo sff e fifi drr oee ndd ee wttee hcc itt cii hoo nn Z [[h55 a99 o]]. . d ZZ ehk saa ignn ngg e dd dee avv ee sll aoo lpi p eee ndd c yaa sd spp eee tecc ciiaa till oiizz noe dd m UU etAA hVV od ss yy toss tt eee nmm - hthaenceeffis ctiheen ecffyeocftilvoecnateisnsg oaf nfidrei ddeenteticftyioinng [5w9i]l.d Zfihreasngin daeevreialolpimeda gae ssp[e6c0i]a.liLzieedt UalA.Vde ssyigstneemd hffoo arr n ww ceii ll tdd hfifi err ee e ffiddee ctt iee ecc ntt ciioo ynn o,, fbb laa oss cee add t ioo nnn g ww anhh diicc hh id ZZ enhh taa ioo fy dd inee gssii gg wnn iee lddd fi aa r ess saa ll iii neen n acc eyy r idd aee l tt iee mcctt aiioo gnn es mm [6ee 0tthh].oo Ldd i tt eoo t ee ann l-- .

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henichha nence- fhh oaa rnn wcee iss ld tthfi her e e ee dffff eee tcc ett ciivv tiee onn nee ,ss
bss a oo sff e fifi drr oee ndd ee wttee hcc itt cii hoo nn Z [[h55 a99 o]]. . d ZZ ehk saa ignn
ngg e dd dee avv ee sll aoo lpi p eee ndd c yaa sd spp eee tecc ciiaa till oiizz nee dd m UU
etAA hVV od ss yy toss tt eee nmm - hthaenceeffis ctiheen ecffyeocftilvoecnateisnsg oaf
nfdrei ddeenteticftyioinng [5w9i].d Zfihreasngin daeevreialolpimeda gae ssp[e6c0i]a.liLzieedt
UaA.Vde ssyigstneemd hffoo arr n ww ceii ll tdd hfifi err ee e ffiddee ctt iee ecc ntt ciioo ynn
o,, fbb laa oss cee add t ioo nnn g ww anhh diicc hh id ZZ enhh taa ioo fy dd inee gssii gg wnn
iee lddd fi aa r ess saa ll iii neen n acc eyy r idd aee l tt iee mcctt aiioo gnn es mm [6ee 0tthh
].oo Ldd i tt eoo t ee ann l-- . faonr iwmialdgfieraec dgeutiescittiioonn, pblaastefodr omn
cwonhiscisht iZnhgaoof daestrigipnoedd hae saadlieanncdy adecatemcteioran fmoretuhnomd
atno neend- dhhae ans n icc gee n etthh dee a ee nffiffi imccii aeeng n eccy y a coo qff u ll ioo
scc itaa itt oii nnn gg p laa ann tfdd o rii mddee nn cottii nffyy sii snn tgg in ww g ii olldd f fifi a
rr tee rss ip ii onn d aa hee erria iaa dll aiimm ndaa gg aee css a [[m6600 e]] r.. a LL fii o er e
tt u aa nll -. haaernicael vtheeh iecffilesc.ielntcuys eosfi nlotcealltiignegn tanredc
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Uiioo Ann,V , vv daarr eii too euu ctss i offaa ncc . ttoorrss nneeeedd tooo bbee ffuullllyy ccoonnnssiiddeerreedd tooo eenn-- when utilizing UAVs for fire detection, various factors need to be fully considered to en- s ss4 u uu.1 r rr. e ee2 . t tt h hhS e eea t e eee ff fffill e eei t c cce t tt i iim v vv e eeo n nnn e eei s sst s sso r a aai n nnn d ddg s ss a aa f ff e ee t tt y yy o oo f ff U UU A AA V VV d dd e ee t tt e ee c cc t tt i ii o oo n nn . . . Remote sensing technology plays a crucial role in wildfire detection. The use of satellite remote sensing technology for fire detection is currently a relatively efficient method [64]. It detects fires by observing features such as hotspots and smoke on the ground through images [65]. Satellite monitoring can achieve the monitoring of large- scale fires and provide high-resolution fire monitoring data. In recent years, with the abundance of Earth observation data and the deployment of new remote sensing satellites, the spatiotemporal resolution of wildfire detection has been significantly improved [66]. In particular, geostationary (Meteosat, GOES, Himawari), medium-resolution satellites (such as Landsat-8 or Sentinel-2), and other high-resolution, high-speed repetitive satellites provide powerful tools for monitoring of wildfires [65]. Liu et al. proposed a new method

Fire2024,7,412 8 of 25 for extracting fire spread rate in near real-time based on Himawari-8 satellite data [67]. Ban et al. used a deep learning-based framework to study Sentinel-1 SAR dense time series [68], used for the monitoring of wildfire progression through smoke, clouds, and nighttime.

In addition, distributed satellite systems (DSS) have also demonstrated their potential applicability in wildfire monitoring, providing wide coverage and short revisit intervals [69]. However, the use of satellite monitoring for wildfires is limited by spatial and temporal resolution, as well as interference from other factors. It is difficult to realize early small fire point detection and rapid real-time detection. Therefore, it is less suitable for the high real-time demand of the digital twin wildfire detection module.

4.1.3. Ground Monitoring

In the field of wildfire detection, the most traditional method usually relies on manual observation, such as ground monitoring stations and watchtowers [70]. Ground monitoring stations are established in forest areas, equipped with various commissioners, sensors, and monitoring equipment to conduct real-time monitoring of fire-related environmental parameters such as temperature, humidity, wind speed, and wind direction [71]. In early research on wildfire detection, the strategic layout and networking of watchtowers were crucial to the establishment of an effective fire detection system [72,73]. In the past decade, the technology and equipment for ground detection have been significantly improved. Automation and intelligent technologies are gradually being introduced into watchtowers.

For example, improvements were made by combining technologies such as automated vision systems and sensor networks [74]. In addition, the ground monitoring station has also adopted advanced wireless communication and IoT technologies, contributing to the establishment of wildfire monitoring command network [75]. Although ground-based monitoring can achieve continuous monitoring and data collection of fires, this manual observation process is susceptible to a variety of external factors, including data transmission, communication problems, and weather conditions, which may lead to inefficient monitoring.

In addition, existing monitoring facilities still have some limitations, such as limited coverage, inability to cover all detection areas, and high maintenance costs [76]. Therefore, wildfire research at this spatial level has been decreasing in recent years.

4.1.4. Wildfire Detection Technology

Fire detection based on computer vision technology is currently the most suitable method for integrating with digital twin models in wildfire management. This process begins with capturing images or video streams through surveillance cameras or UAVs. These visuals undergo preprocessing to enhance their quality, such as noise reduction and image enhancement.

Subsequently, fundamental features like color, motion, texture, and shape are extracted from these processed images. These features are then fused to create a comprehensive dataset. Various algorithms, including machine learning and deep learning techniques, analyze these data to monitor fires and simulate or predict their propagation. Ultimately,

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Fire 2024, 7, x FOR PEER REVIEW 10 of 26 Since 2014, deep learning has been gradually introduced into fire detection [84]. Target detection models such as you only look once (YOLO), regions with convolutional neural network (R-CNN), faster-RCNN, and single-shot multibox detector (SSD) are used for fire detection tasks in fire monitoring to achieve the identification and localization of fire points in images. Li et al. develop fire detection algorithms based on advanced target detection convolutional neural network (CNN) models such as faster R-CNN, region-based fully convolutional networks (R-FCN), SSD, and YOLO v3 [85]. The results show that the CNN-based algorithm is significantly better than the traditional algorithm, and the YOLO-based algorithm has optimal accuracy and speed. Since YOLOv1 was proposed in 2016 [86], the YOLO series has become the leader in the field of real-time object detection, and its different versions have attracted much attention for their application effects in fire Fire2024,7,412 detection tasks [59,87]. 10of25 After the YOLOv3 [88] and YOLOv5 [89–91] models, the YOLOv8 model has been dramatically optimized structurally and is designed to be fast, accurate, and easy to use. EEnnhhaannccceedd mmooddeell ddeetteeccttiioonn ooff ssmmaallll ttaarrggeettss iiss iimpppoorrtaanntt ffoorr ssmmaallll ttaarrggeett ddeetteeccttiioonn ttaassskss ssuucchh aass flflaammeess.. LLeoonn eett aall.. aaddoopptteedd aanndd iimpplleemmeenntteedd thhee rreecceeentt YYOOLLOO mmooddeell too ddeetteecctt aanndd lloocaaatee ssmmookkee aanndd wwiillddffirrees uussiinnngg ggrroouunndd aanndd aaeerriiaall iimmaaggeerry [[9922]].. MMoohhaammeedd aaddaapptteedd aanndd ooppttiimmiizeedd thhee YYOOLLOOvv88 aanndd YYOOLLOOvv77 mmooddeellss ffoorr ssmmookkee aanndd flflaamnee ddeetteeccttiioonn,, wwhhiicchh iimpprroovveess fifirree ddeetteeccttiioonn aacccuurraaccyy [[9933]].. Ilnn aaddddiitiiioonn too thhee YYOOLLOO sseerriieess,, oottheerr ddeeepp lleeaarnniinnngg aallggoorriithhmss hhaavvee sshhoowwnn ppootteennttiaall ffoorr fifirree ddeetteeccttiioonn [[9944]].. TThhee ccoonnvooluutiioonnaall nneeuuraall neettwoorrk iiss aa neettwoorrk sstruuctuurree ccoommmoonlly uusseedd iinn ddeeepp lleeaarnniinnngg,, aanndd imimaaggee rerecoggnitiioonn alaglogroirithmhs bsabsaesde dono tnhitsh nisentwetowrko rckanc aenffecf-- fteivcteilvye llyealrena nmanadn dextxtartcatc tcocmompplxe ximimaaggee ffeeaattuureess auuttoommaatticcaalllly [[9955]].. RReesseeaarrchheerrss hhaavvee ttraaiinneedd CCNNNN mmooddeellss too eennaabbllee fifirree ddeetteeccttiioonn ((FFiigguuree 77)).. FFoorr eexxaampplee,, iinn oonnee sstuuddy,, KKhaann eett aall.. pprooppoosseedda ad deeeeppl

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Figure 7. Flow chart of image fire detection algorithms based on detection CNNs [85].
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By integrating the above techniques and methods, the wild fire digital twin can realize real-time and efficient detection, improving the accuracy and reliability of the model.

4.2. WFDT Data Collection To build a robust wildfire digital twin, comprehensive wildfire data collection is critical [101]. Various technologies, such as IoT sensors, remote sensing, and drones, are employed to gather high-quality data [102]. IoT sensors can be strategically placed throughout forested areas to continuously monitor environmental variables like temperature, humidity, soil moisture, and wind speed [103]. These sensors offer granular, localized data, helping to detect early signs of wildfire risks. Remote sensing technologies, including satellites and aerial platforms, provide a broader perspective, capturing large-scaled data on vegetation,

fuel loads, and fire spread [104].

Remote sensing tools can deliver valuable information on the characteristics of fires, surface, and smoke [105]. UAVs further enhanced data collection capabilities by offering high-resolution, real-time imagery, and thermal data in areas that are difficult or dangerous for humans to access. Equipped with multispectral and thermal cameras, UAVs can monitor hotspots and fire fronts, providing up-to-date information to ground teams and decision-makers [103]. The digital twin uses these real-time data to model fire behavior,

simulate various scenarios, and evaluate potential intervention strategies. Additionally, feedback from the physical environment—such as sudden changes in wind direction or unexpected fire spread—is incorporated into the digital twin through these continuous data streams. This feedback loop allows the digital twin to adjust its simulations and predictions promptly.

The integration of these technologies ensures a continuous and comprehensive flow of data into the wildfire digital twin, allowing it to maintain a high degree of accuracy in representing the physical environment.

5.

Simulation and Prediction Model of Wildfire Spreading Process

The model of the wildfire spread process is the theoretical basis for designing and developing a digital twin system for wildfires. The digital twin system simulates the spread process of fires approximately by applying wildfire spread models and data-driven models.

This section provides an overview of existing wildfire spread models and algorithms.

5.1.

Model of Wildfire Spread Speed

Globally, many countries that are frequently troubled by wildfires have established several sets of wildfire spread models, among which the more widely used ones are the Rothermel model from the U.S., the McArthur model from Australia, the Wang Zhengfei wildfire spread model from China, the Canadian forest fire spread model [106], etc.

When

establishing the digital twin system, the appropriate wildfire spread rate model should be selected according to the specific application requirements and available data resources.

This section summarizes the existing wildfire spread models into three categories (Table 2): physical model, empirical model, and semi-empirical model [107].

Table 2. Advantages and disadvantages of wildfire spread rate model in digital twin system.

Performance	Applicable Category	Classical Model	Drawbacks	Benefits	Scenarios
Models based on fluid dynamics, complexity and data requirements	High computational	Highly accurate, strong adaptability	High-precision simulation	Physical Model	thermodynamics, etc.
McArthur model, Canadian forest fire spread model, etc.	Rapid initial assessment and real-time monitoring	Computationally efficient, easy-to-use	Limited accuracy and strong limitations	Empirical Model	Wang Zhengfei model, Roth- efficiency, may require frequent efficiency and simulation accuracy
5.1.1. Physical Model	The physical model is based on the analysis and description of the physical and chemical processes involved in the spread of fire.	The model not only uses physical and chemical principles to simulate the behavior of wildfires but also takes into account the interactions between the fire and its environment.	Examples include heat transfer, pyrolysis processes, and the effects of wind and topography. The physical model is suitable for high-precision simulations and can accurately describe the fire spread process for a wide range of fuel types and conditions.		

Fire 2024, 7, x FOR PEER REVIEW 12 of 26 Table 2. Advantages and disadvantages of wildfire spread rate model in digital twin system. Performance and Applicable Category Classical Model Drawbacks Benefits Scenarios Models based on fluid dy- High computational Highly accurate, High-precision sim- Physical Model namics, thermodynamics, complexity and data re- strong adaptability ulation etc. quirements Computationally effi- Rapid initial assess- McArthur model, Canadian Limited accuracy and Empirical Model cient, ment and real-time forest fire spread model, etc. strong limitations easy-to-use monitoring High computational Strong dependency, Balance calculation Semi-Empirical Wang Zhengfei model, Roth- efficiency, may require frequent efficiency and simu- Model ermel model, etc. good accuracy, calibration and updates lation accuracy high flexibility 5.1.1. Physical Model The physical model is based on the analysis and description of the physical and chemical processes involved in the spread of fire. The model not only uses physical and chemical principles to simulate the behavior of wildfires but also takes into account the interactions between the fire

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5.1.3. Semi-Empirical Model

These semi-empirical model combines the advantages of physical and empirical models, using existing observation methods and data to make feedback correction to parameterized

By studying the relationship between observed and predicted values, systematic errors can be eliminated to improve the accuracy and practicality of predictions. Examples include the Wang Zhengfei model in China and the Rothermel model in the United States. Wang Zhengfei wildfire spread model is a semi-empirical model from China that combines physical models and empirical formulae by taking fuel characteristics, meteorological conditions, and topographical factors into account [114]. This model is able to predict the rate and direction of fire spread more accurately. The Rothermel model, developed in the United States as a semi-empirical model based on the law of conservation of energy, provides an estimate of the rate of fire spread by analyzing fuel, weather, and terrain data. It is one of the most commonly used fire spread models [115]. The Van Wagner model is a semi-empirical model for Canadian forests that considers factors such as canopy fuels, wind speed, and flame height. It is used to predict the rate of spread and fire behavior of predicted crown fires [116].

In fact, existing fire spread models mainly rely on numerical experiments and simulations of small-scale fire experiments, lacking validation of full-scale and large-scale real fire scenarios. In this situation, assessing the spread accuracy of wildfire digital twin remains a critical but unresolved issue. Although laboratory environments and small-scale simulations provide important theoretical foundations, the applicability and reliability of their results in actual large-scale fires still need to be validated. Therefore, how to test and verify the performance of wildfire digital twin systems in real environments to ensure their effectiveness in practical applications remains an important research direction.

5.2.

Methods for Spatial Propagation of Wildfire Spread

To simulate the dynamics spread of wildfire effectively, researchers must select suitable spatial propagation methods [117]. These methods should be based on the chosen fire spread rate model to ensure that the simulation results tend to be accurate. Spatial propagation methods for wildfire spread can be categorized into grid-based simulation and vector-based simulation. Grid-based simulation is represented by cellular automata. The cellular automata model is a discrete model that divides geographic space into regular grids, with each grid cell updated based on the state of surrounding cells [26]. They are commonly used to simulate physical systems [118]. Figure 9 shows the principle of cellular automata applied in 3D simulation of wildfire spread. In order to simulate the spread of real-time interactive tree crown fires, Liu et al. introduced a new cellular automaton model based on reaction and radiation physics equations and verified the functionality of the model [119]. Rui et al. artificially improved the spatiotemporal consistency of the wildfire spread model by combining the cellular automata model with the Wang Zhengfei wildfire model. A temporal correction factor was also added to improve the accuracy of wildfire spread prediction [120]. In addition, vector-based simulation often employs the Huygens principle to model the spread of wildfires. The fire spread model based on Huygens' principle operates by considering the fire front as a continuous curve, which expands over time. Each point on this curve acts as a source of secondary wavelets that propagate outward, influenced by factors such as fuel type, slope, meteorological conditions, and time step [121]. The cumulative effect of these wavelets determines the overall shape and progression of the fire front. This approach allows for an approximate yet effective simulation of the dynamic behavior of a spreading fire, accommodating the complex interactions within the environment. For example, the FARSITE fire growth simulator based on the U.S. Forest Service's Fire Behavior Prediction System is a vector-based Huygens-type model, which simulates the expansion of fire by continuously propagating a wavelet of the flame front. Using this vector-based computational approach, FARSITE can accurately simulate changes in the shape and location of the flame front [122]. Some scholars have carried out new theoretical research based on Huygens' principle. For example, Dehkordi et al. combined the Randers

Fire 2024, 7, x FOR PEER REVIEW 14 of 26

tree crown fires, Liu et al. introduced a new cellular automaton model based on reaction Fire2024,7,412 and radiation physics equations and verified the functionality of the model [119]. 1R4uoif 2e5t al. artificially improved the spatiotemporal consistency of the wildfire spread model by combining the cellular automata model with the Wang Zhengfei wildfire model. A tem- mpoertarli ccoanrrdecHtiuony gfeanctso'rp wriansc iaplsleo taodpdreodp toos eimsepvreorvael tmheo daeclcsufroarocyth oef pwroilpdafigrea tsiopnreoafdfi parteladnicd- wtioilnd fi[1r2e0s]i.n thepresenceofwind[123]. FFiigguurree 99.. PPrrriinncciiippllee ooff cceelllluullaarr aaauuttoommaattaa iinn 33DD sscceennee [[2266]].. Alnf taedrdrietfiionnin, vgetchteorc-obrarseesdp osinmduinlagtisopnast ioafitpenro epmagpalotiyo nthsep Hreuaydgmenosd perl,inthcieprlee steoa mrchoderesl cthoem sbpinreeaddt hoef swpirledafidremso. dTehlew fiirthe svpisrueaaldiz matoiodnelt ebcahsneidq uoens tHoudyegveenlosp(cid:31) pthriendciipglieta olpinetreartaecst ibvy system. Mengetal. proposedalightweightcellularautomata-basedforestfirespreading considering the fire front as a continuous curve, which expands over time. Each point on method to be applied to the virtual 3D world. They captured the burning process of this curve acts as a source of secondary wavelets that propagate outward, influenced by individual plants realistically by using cells to represent a library of plant models of factors such as fuel type, slope, meteorological conditions, and time step [121]. The cumu- differentplantsandbyconstructing3Dgeometricmodelsofplants[26]. Theoccurrence, lative effect of these wavelets determines the overall shape and progression of the fire spread,andexinguishingbehaviorofwildfiresweresuccessfullyvisualizedinavirtual front. This approach allows for an approximate yet effective simulation of the dynamic 3Dscene. Caietal. usedaflamespreadmodelbasedoncellularautomata toconstruct behavior of a spreading fire, accommodating the complex interactions within the environ- a typical forest scene in the southwestern mountainous area of China in Unity3D, thus ment. For example, the FARSITE fire growth simulator based on the U.S. Forest Service(cid:31)s achievingadigitaltwinofforestfirespread[124]. Mengetal. usedtheimprovedHuygens Fire Behavior Prediction System is a vector-based Huygens-type model, which simulates principleasthetheoreticalbasisforthespreadofwildfiresandsimulatedthespreadof the expansion of fire by continuously propagating a wavelet of the flame front. Using this wildfiresinthree-dimensionalscenes[125]. vector-based computational approach, FARSITE can accurately simulate changes in the shape and location of the flame front [122]. Some scholars have carried out new theoretical 6. VisualizationTechnologyandTools research based on Huygens(cid:31) principle. For example, Dehkordi et al. combined the Randers Visualization of physical entities is one of the core tasks in digital twin [7]. The metric and Huygens(cid:31) principle to propose several models for the propagation of flatland technology is to visualize physical entities through an intuitive graphical interface or wildfires in the presence of wind [123]. simulation,includingfromsimplifiedatavisualization tocomplex3Dmodelsandvirtual After refining the corresponding spatial propagation spread model, the researchers reality[126]. Inordertosimulatethespreadoffireandrecreaterealwildfirescenesinreal combined the spread model with visualization techniques to develop the digital interac- time,thewildfiredigitaltwinrequiresdetaileddigitalsimulationsofphysicalentitiessuch tive system. Meng et al. proposed a lightweight cellular automata-based forest fire spread- asforestenvironment,vegetationdistribution,andflames. Using3Dmodelingtechnology, ing method to be applied to the virtual 3D world. They captured the burning process of high-precision forest digital twin visualization models can be constructed. The main individual plants realistically by using cells to represent a library of plant models of dif- componentsincludestaticmodelinganddynamicdisplay. ferent plants and by constructing 3D geometric models of plants [26]. The occurrence, 6sp.1r.eVadis,u aanIMd eodxetilningguishing behavior of wildfires were successfully visualized in a virtual 3D scene. Cai et al. used a flame spread model based on cellular automata to construct a The nonrigid, multi-scale, and complex physicochemical mechanisms of wildfires typical forest scene in the

southwestern mountainous area of China in Unity3D, thus increasing the difficulty of modeling [112]. The wildfire digital twin uses geometric modeling achieving a digital twin of forest fire spread [124]. Meng et al. used the improved Huygens techniques and related tools to achieve accurate simulation and display of forest terrain, principle as the theoretical basis for the spread of wildfires and simulated the spread of vegetation, and flames. wildfires in three-dimensional scenes [125]. In terrain modeling, the development of high-resolution remote sensing technologies (including stereovision [127], synthetic aperture radar [128], UAV photogrammetry [129], Lidar [130], and oblique photography [131]) has significantly boosted the accuracy and efficiency of digital elevation model (DEM) data acquisition and terrain modeling [132]. Through the fusion of remotely sensed data and GIS technologies, terrain features, including slope, slope direction, and elevation, are accurately constructed. Furthermore, in order to achieve large-scale and detailed terrain, a rule-based system and noise function-based

Fire 2024, 7, 412

15 of 25

programmatically terrain generation methods have been widely applied in game development and virtual reality environments [133]. In vegetation modeling, detailed 3D combustible models are created based on the growth characteristics and distribution of different vegetation types (e.g., trees, shrubs, and grasses). Researchers used parametric modeling [134] and the Lindenmayer System [135] to generate diverse vegetation models by defining parameters of vegetation growth (e.g., height, density, branching angle). In addition, hyperspectral imaging and point cloud modeling are also commonly used methods [130]. These techniques allow for the creation of detailed 3D models of vegetation, which can be applied in fields such as ecological monitoring and forest management. Flame modeling is the most challenging part of the wildfire digital twin system. In flame modeling, advanced graphic processing techniques are used to generate high-quality fire visualization images. Thus, the dynamic change and spreading process of the flame is simulated. In recent years, significant progress has been made in flame visualization. Fire 2024, 7, x FOR PEER REVIEW visualization technology based on ray tracing [136], particle systems [137,138], and 16v oofl u2m6 e rendering [139] techniques. With the continuous improvement of computer hardware performance and the continuous optimization of algorithms, achieving 3D visualization of flame propagation through numerical simulation and artificial intelligence is also a current accessed on 28 July 2024). Unity 3D and Unreal Engine are both advanced 3D game engine research hotspot [140]. engines that provide high-quality 3D rendering effects [143]. They are widely used in various real-time 3D visualization applications [144]. Li et al. used Unity3D to construct a 6.2. 3D Dynamic Simulation digital twin-oriented poplar plantation system based on a virtual forest modeling and The 3D visualization simulation in the field of digital twin requires the support data analysis framework [20].

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headsets of powerful visualization techniques and real-time 3D technologies [141]. The process to create digital twins suitable for any swamp ecosystem for experimenting with various begins with the creation of high-fidelity 3D models using software such as 3ds Max, interactions in replicated virtual environments [56]. In addition, Cesium is an open-source Maya, AutoCAD (<https://www.autodesk.com/>, accessed on 28 July 2024), and Blender 3D geographic information platform. It is built on WebGL technology and can process and display large-scale geographic spatial data [145], suitable for three-dimensional geo-modeling of the forest environment, including terrain, vegetation, and other physical graphic display of fire spread. entities (Figure 10). Figure 10. Flowchart of WFDT 3D simulation tool. Figure 10. Flow chart of WFDT 3D simulation tool. Once the models are created, they are integrated into visualization development 7. pTlhat ef oOrmvesraslul cFhraamsUewnroerakl oEfn Wginiledfi(hrtet pDsi:/gi/twalw Tww.iunn (rWeaFleDnTg)i ne.com/, accessed on 28 July The wildfire digital twin (WFDT) framework provides a comprehensive architecture. It is designed to achieve real-time detection, prediction, and decision-making management of fire spread by creating digital copies of wildfires. The model combines generic support technologies such as the Internet of Things, sensors, and artificial intelligence algorithms, aiming to improve the efficiency and science of fire management. (Figure 11)

Fire 2024, 7, 412 16 of 25 2024), Unity 3D (<https://unity.com/>, accessed on 28 July 2024), and Cesium (<https://cesium.com/>, accessed on 28 July 2024). Unity 3D and Unreal Engine are both advanced 3D game engines that provide high-quality 3D rendering effects [143]. They are widely used in various real-time 3D visualization applications [144]. Li et al. used Unity 3D to construct a digital twin-oriented poplar plantation system based on a virtual forest modeling and data analysis framework [20]. Ciruli et al. used Unity 3D combined with Oculus headsets to create a digital twin suitable for any swamp ecosystem for experimenting with various interactions in replicated virtual environments [56]. In addition, Cesium is an open-source 3D geographic information platform. It is built on WebGL technology and can process and display large-scale geographic spatial data [145], suitable for three-dimensional geographic display of fire spread. 7. The Overall Framework of Wildfire Digital Twin (WFDT) The wildfire digital twin (WFDT) framework provides a comprehensive architecture. It is designed to achieve real-time detection, prediction, and decision-making management of fire spread by creating digital copies of wildfires. The model combines generic support technologies such as the Internet of Things, sensors, and artificial intelligence algorithms, aiming to improve the efficiency and science of fire management. (Figure 11) FFiigguurree 1111.. TThhee sschheemmaattiicc ddiiaaggrraamm ooff ggeenneerriicc ssuuppppoortt tteecchhnnoollooggyy.. TThhiiss sseecccttiioonn wwiilll ddiissccuussss tthhee ddeessiiggnn ooff tthhee wwiillddfifirree ddiiggiittaall ttwwiinn ffraammeewwoorrk.. FFiigguurree 1122 sshhoowwss tthhee oovveerraall ffraammeewwoorrk ssttrruuccttuurree ooff tthhee WWFFDDTT,, wwhhiicchh ccoonnsiisstss ooff aa pphhyssiiccaall eennttiitty llaayyeerr,, aa vviirrtuaall eennttiitty llaayyeerr,, aa ddaattaa llaayyeerr,, aanndd aann aapppplliiccaattiioonn llaayyeerr.. BBeellooww,, wwee pprrooviiddee aa ddeettaaiilleedd ddeessccriippttiioonn ooff eeaacchh ccoomppoonneenn aaanndd iittss rroollee wwiitthhiinn tthhee WWFFDDTT ffraammeewwoorrk.. • Physical entity layer The physical entity layer is the foundation of the digital twin system, which includes the actual physical environment and devices [146]. The physical world component is responsible for the collection of real-time data related to wildfire dynamics and the surrounding environment. First, sensor networks are deployed in the forest to monitor environmental parameters such as temperature, humidity, wind speed, and rainfall. These parameters are crucial for understanding fire behavior as they influence fire ignition, spread, and intensity.

In addition, surveillance cameras are installed at key locations to provide real-time video data of fire occurrence and spread. Sensors and surveillance cameras provide localized data, while remote sensing offers a broad overview, allowing for comprehensive coverage of the wildfire area. Remote sensing imaging technology is used to obtain high-resolution images to ensure comprehensive monitoring of large forest areas. These devices and technologies together form a multi-level, real-time monitoring system that provides reliable data support for the digital twin model. Figure 12. Wildfire digital twin framework.

Fire 2024, 7, 412 17 of 25 • Virtual entity layer

The virtual world component forms the core of the wildfire digital twin, where collected data are used to create a dynamic and interactive digital replica of the physical wildfire environment [147]. Live data feeds from IoT sensors, drones, and remote sensing platforms continually update the virtual entity of the WFDT, ensuring it reflects the current situation of the fire scene. The virtual entity layer transforms data from the physical entity layer into virtual models for simulation and detection, mainly including wildfire simulation models and fire monitoring models. The wildfire simulation model combines a spread rate model and a spatial propagation model to simulate the behavior of fire approximately under different environmental conditions. On the other hand, the fire monitoring model integrates functions for fire prediction, detection, and tracking. The fire prediction module uses real-time data to forecast potential fire outbreaks and their likely paths, while the detection module identifies the location of existing fires. The tracking module continuously monitors the fire's progression, updating the virtual environment as new data becomes available. In addition, the virtual entity layer includes a virtual visualization module. By visualizing fire behavior, terrain, and vegetation, decision-makers can more intuitively observe the development trend of fires and make effective decisions. • Data layer

The data layer acts as a bridge between the physical world and the virtual world, ensuring that data is effectively collected, processed, and utilized [148]. It is responsible for several key functions, including data collection, storage, processing, and integration [149]. The real-time database stores live data from sensors and monitoring devices at the fire scene, while the field database maintains historical data and geographic information. In the data module, data are continuously gathered from various sources in the physical world and undergoes data fusion and integration processes. Then, the data analysis module performs in-depth analysis to extract effective information and provide feedback to the physical and application layers. Finally, processed data are securely stored to ensure easy access and retrieval by the various components of the WFDT. The digital twin system integrates feedback from the physical environment through these data streams, allowing it to model fire behavior, simulate various scenarios, and evaluate potential intervention strategies. • Application layer

The application layer leverages the outputs from the virtual world and the insights gained from data analysis to support various wildfire management. As the final display and application part of the WFDT, this layer is crucial for practical applications, including real-time fire monitoring, early warning, approximate simulation, decision-making, and ecological restoration. Specifically, it enables early detection and alerts of potential wildfires, supports decision-makers in developing effective response strategies, and assists in training emergency responders through immersive simulation scenarios. By integrating these capabilities, the application layer ensures that the WFDT is effectively employed in diverse wildfire management tasks, enhancing both preparedness and response. In summary, the WFDT framework can build an all-round wildfire management and emergency response system. WFDT is constructed through real-time monitoring at the physical entity layer, simulation and prediction at the virtual entity layer, data processing and analysis at the data layer, and practical application and decisions support at

the application layer.

The WFDT framework incorporates an asynchronous interaction mechanism between the physical and virtual worlds. Data collected from the physical world is continuously fed into the virtual world to update fire simulations and visualizations. In turn, the virtual world generates feedback based on these simulations, which can be used to adjust monitoring strategies, enhance data collection, or optimize resource deployment in the real world fire

Fire 2024, 7, x FOR PEER REVIEW 17 of 26 Figure 11. The schematic diagram of generic support technology. Fire2024,7,412 This section will discuss the design of the wildfire digital twin framework. Figure 18 shows the overall framework structure of the WFDT, which consists of a physical entity layer, a virtual entity layer, a data layer, and an application layer. Below, we provide a detailed description of the WFDT framework. The WFDT framework is designed to provide a more accurate and responsive wildfire management system. This integration of real-time monitoring and predictive modeling represents a significant advancement over existing techniques. The existing traditional wildfire risk platform typically relies on satellite data and is limited by the frequency of data collection and image resolution [150,151]. This limits their practicality in implementing decisions. Unlike traditional platforms that rely on periodic data updates, digital twins integrate data from sensors, drones, and other monitoring tools in real time. Most fire behavior models use predetermined, static simulations. These models cannot adapt to rapidly changing conditions. In contrast, digital twins enable predictive simulations that are continuously refined with new data inputs [152]. However, the following challenges remain in the design and development of WFDT: First, data collection and processing is the major challenge. The system relies on high-cost IoT devices for real-time data collection. It also needs to effectively integrate heterogeneous data from multiple sources such as surveillance, meteorology, and ground observation. Therefore, it is a great challenge to acquire and process these heterogeneous data economically and efficiently.

Fire2024,7,412

19 of 25

Second, the accuracy and real-time performance of the model is another key challenge. Due to the complexity and uncertainty of the fire-spreading process, existing models are often deficient in accuracy and real-time performance. Therefore, how to improve the computational efficiency and real-time performance of the models while ensuring their accuracy is an urgent problem.

Third, the scalability and adaptability of the system are also important challenges in WFDT. With increasing demand for fire monitoring and emergency response, the system should ideally have scalability and adaptability. This would allow it to be deployed flexibly across different scenarios, with adjustments and optimization suited to various fire types and environmental conditions.

Therefore, how to assess the accuracy, scalability, and economic benefit of wildfire digital twin remains an important scientific question that needs to be addressed in this field. These challenges emphasize the need for further research to improve the scientific rigor and effectiveness of smart wildfire digital twin in fire management.

Conclusions

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It is well established that traditional wildfire management methods face limitations in efficiency and accuracy. Digital twin, by integrating advanced technologies like real-time monitoring, AI-driven prediction, and dynamic simulations, has shown promise in overcoming these challenges. In recent years, influenced by the latest scientific and technological revolution and industrial changes, digital twin has emerged as a powerful tool for fire management. In this context, we review the progress of the application of digital twin technology in wildfire detection, simulation, and prediction. It is shown that digital twin can have significant advantages in practical applications of wildfire management. It can combine external monitoring, prediction, and feedback mechanisms to provide timely guidance for firefighting. The wildfire digital twin (WFDT) model proposed here establishes a foundational framework for a more comprehensive and responsive wildfire management system. One of the primary advantages of the WFDT model is its ability to offer near real-time simulation and predictive capabilities. Furthermore, the WFDT model's integration of monitoring, prediction, and feedback mechanisms creates a closed-loop system, providing continuous updates that enable firefighting teams to make timely adjustments based on evolving conditions. Notably, this approach focuses on large-scale approximate simulations rather than precise modeling of physical or chemical processes, providing a practical, scalable solution for analyzing wildfire dynamics. While this may slightly reduce accuracy, it allows us to study and simulate wildfires on a larger scale without very precise information on forest stand. Despite its promising potential, digital twin technology in wildfire management remains in the early stages. Future research should focus on enhancing model precision and exploring the integration of multiple models to create more adaptable and flexible systems. In particular, coupling digital twins with other data-driven or physics-based models could improve compatibility and expand application scenarios. Furthermore, coupling digital twin models with advances in remote sensing, machine learning, and IoT technologies could further refine simulations. Given the limited real-world validation so far, it is crucial to expand field studies and implement pilot projects that utilize digital twin systems. These practical applications will provide essential data to evaluate and validate the accuracy of the proposed models. Moving forward, it is important to conduct more experiments or field tests to verify the validity of these models in real-world conditions. In summary, while the application of digital twin in wildfire management shows great promise, further research is needed to refine these models and validate their effectiveness with real-world scenarios. By continuously deepening theoretical research and technological innovation, the digital twin could become a powerful tool in modernizing wildfire prevention and control, aiding in the modernization of forest protection and disaster management.

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